NODE ANALYSIS METHOD

CIRCUIT ANALYSIS METHOD

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

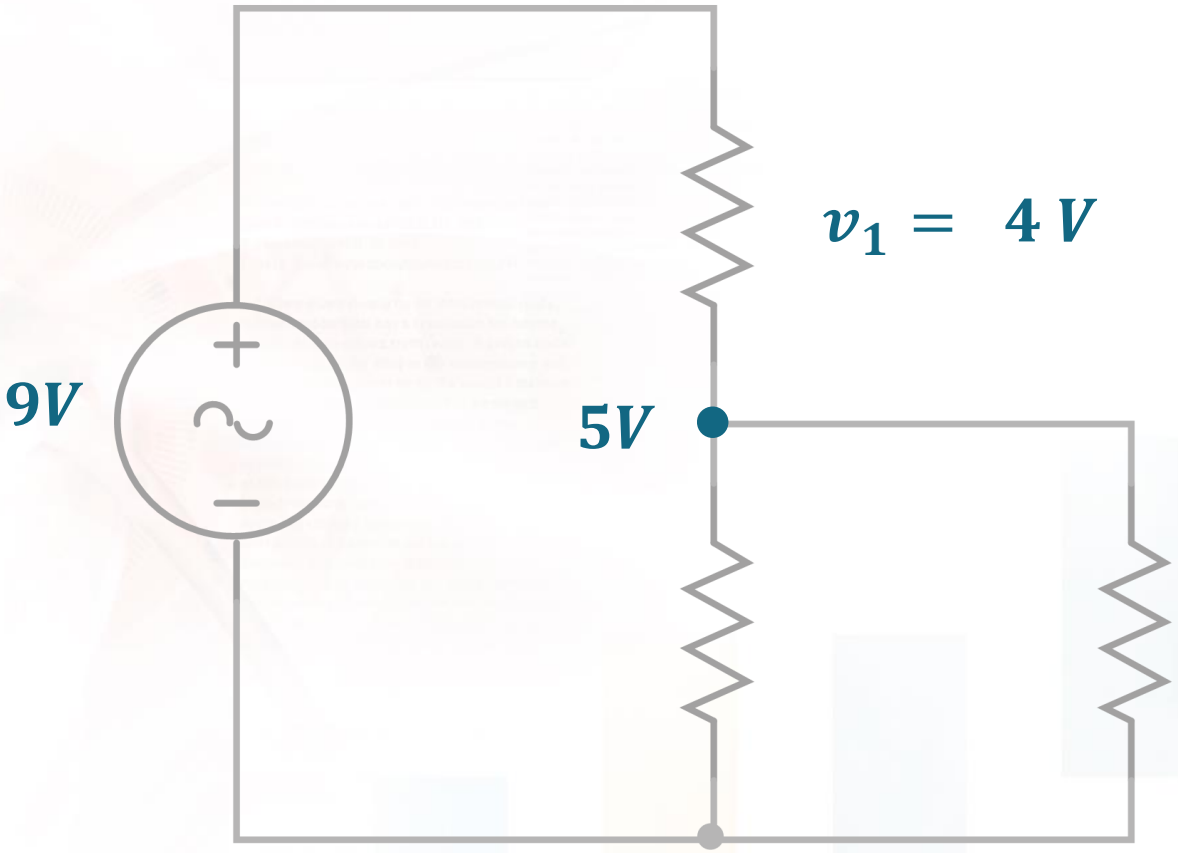
Node Voltage

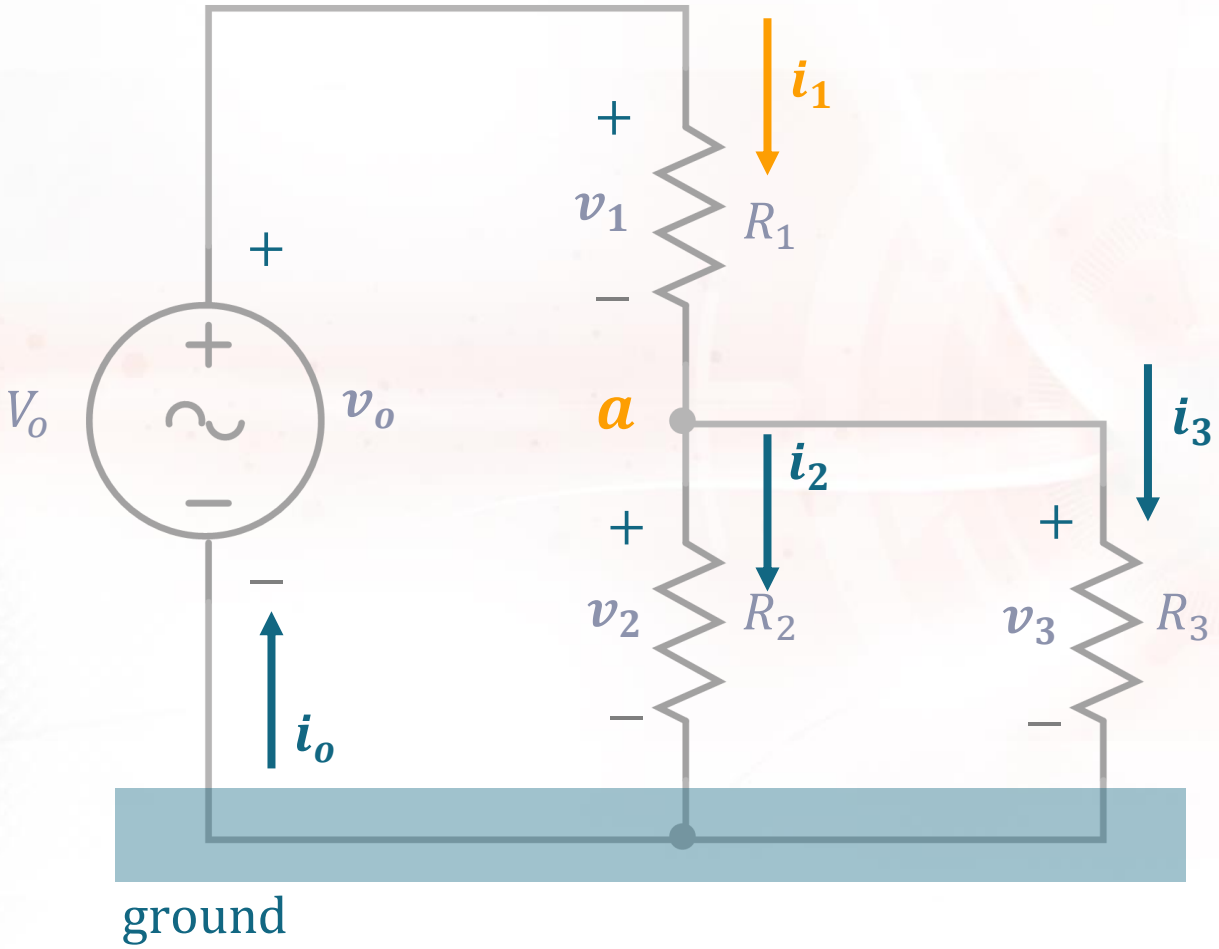
Node Analysis Method

NODE VOLTAGE

NODE VOLTAGE

The voltage **difference** between two points, or **nodes**, in an electrical circuit

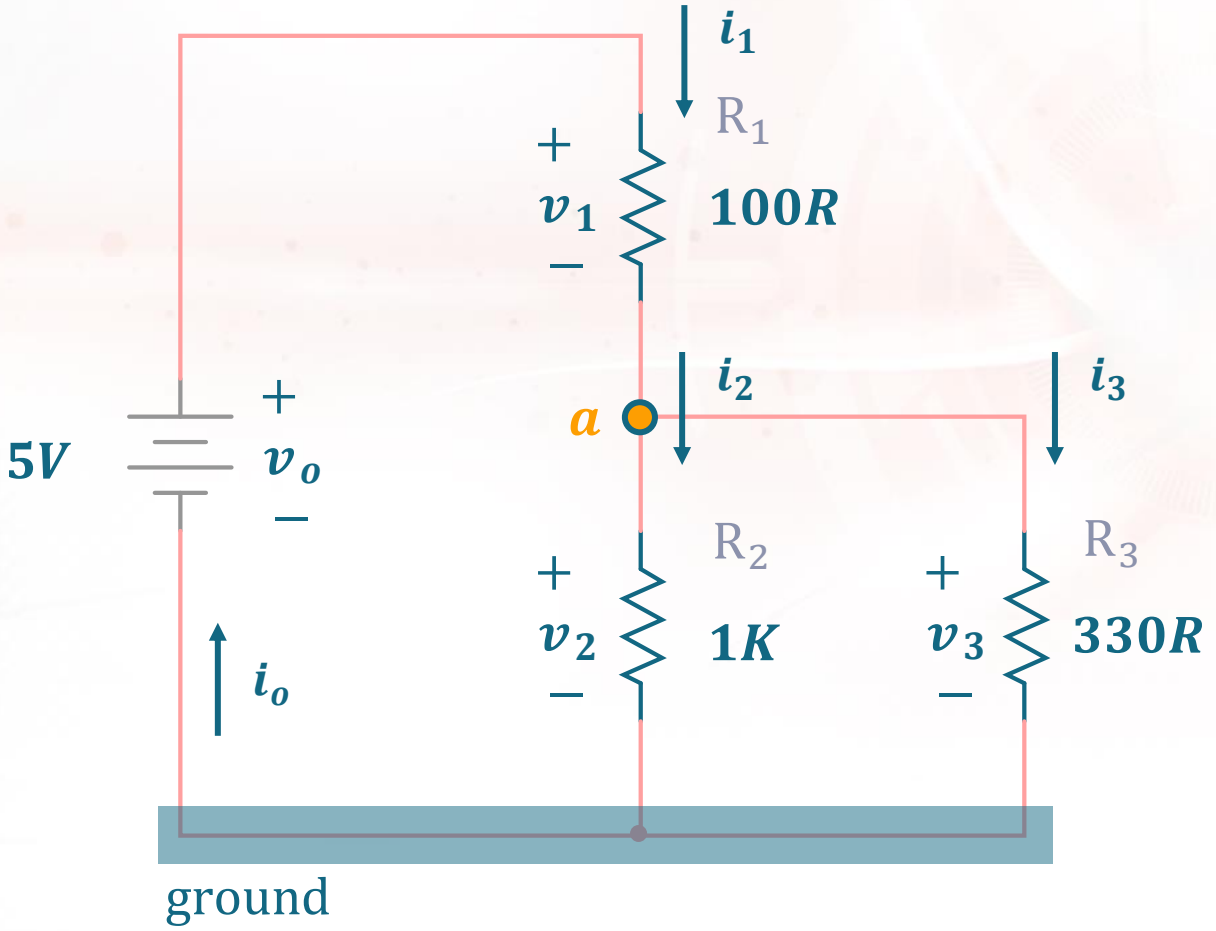




Uses **KCL** implicitly to formula ***vi*** relationship equation

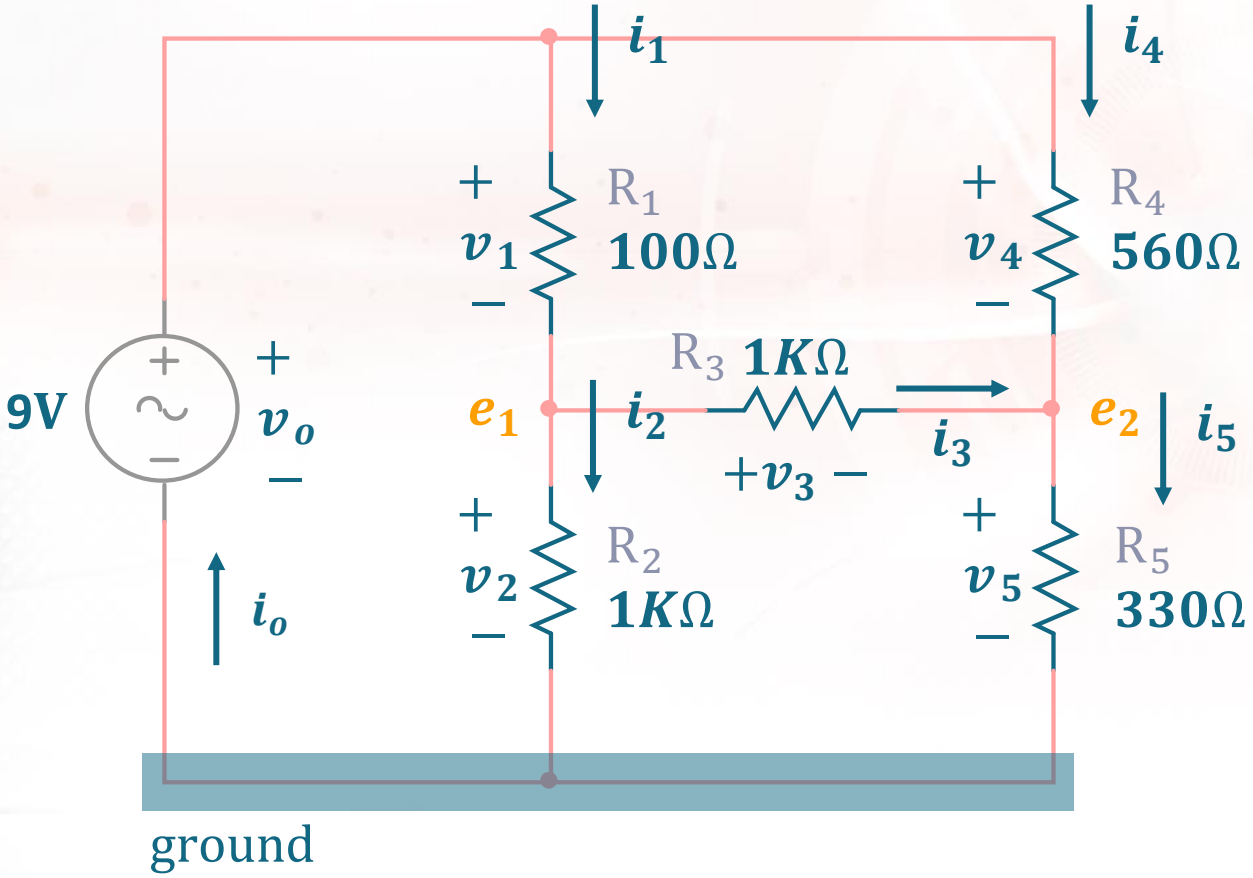
KCL @ *a*,

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



KCL @ a,

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.

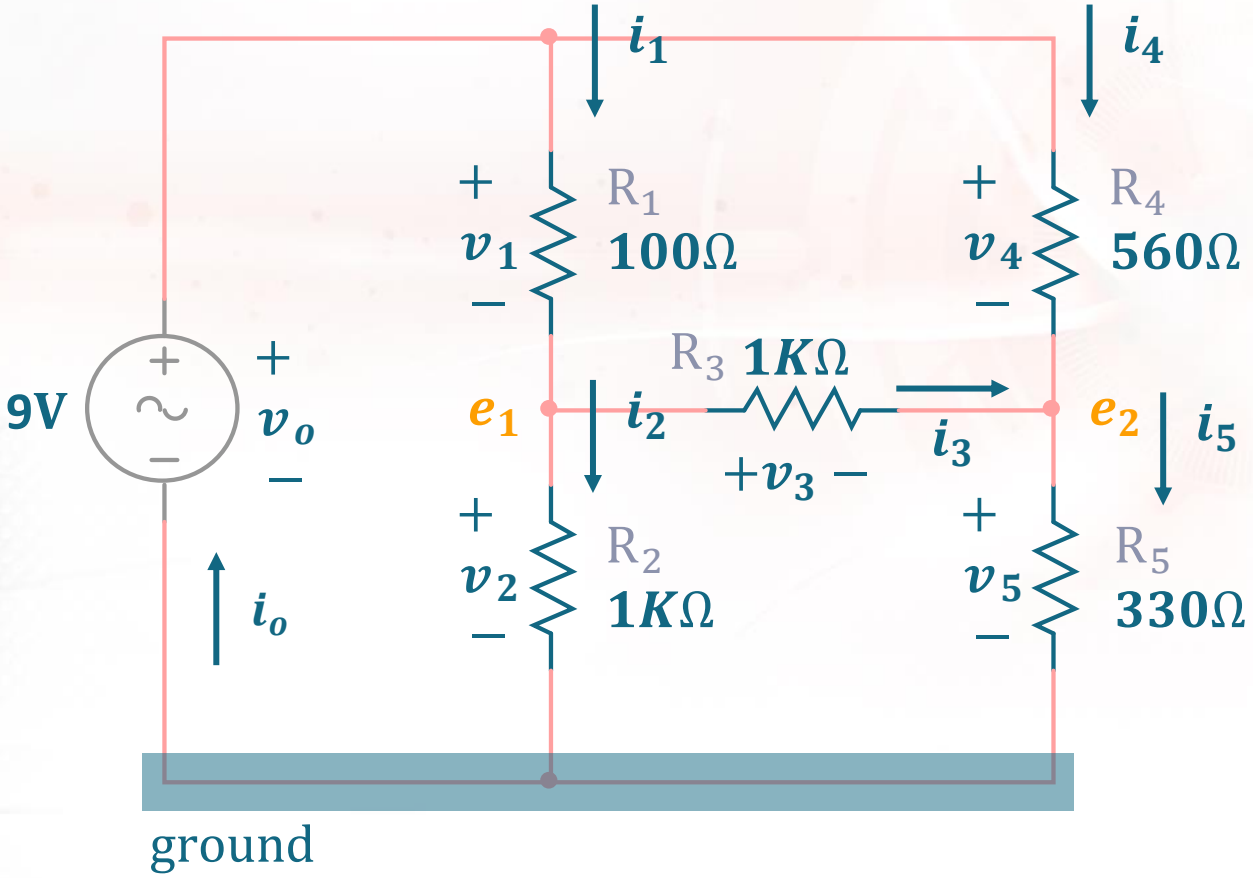


Gaussian Elimination,

Augmented Matrix

$$\left|\begin{array}{cc} e_1 & e_2 \\ & \\ & \end{array}\right| \begin{array}{c} \\ r_1 \\ \\ r_2 \end{array}$$

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



LABORATORY